A-A-50183
October 19, 1988

COMMERCIAL ITEM DESCRIPTION

ICE CREAM MAKER; SHAKE MAKER; AND COMBINATION ICE CREAM AND SHAKE MAKER, SOFT SERVE, MECHANICALLY REFRIGERATED

The General Services Administration has authorized the use of this Commercial Item Description in preference to MIL-I-43705.

1. Classification. This commercial item description covers air cooled soft serve ice cream makers, shake makers, and combination ice cream and shake makers. These machines shall be of the following types, sizes, and styles, as specified (see 5.2).

Type I - Ice cream makers

- Size 1 - 7 gal. per hr. minimum (style A only)
- Size 2 - 12 gal. per hr. minimum
- Size 3 - 30 gal. per hr. minimum, twin head, 15 gal. per hr. each head

Type II - Shake makers

- Size 1 - 15 gal. per hr. minimum (style A only)
- Size 2 - 40 gal. per hr. minimum
- Size 3 - 64 gal. per hr. minimum - twin head, 32 gal. per hr. each head

Type III - Combination ice cream maker and shake maker

- Size 1 - 7 gal per hr. soft serve and 30 gal. per hr. shake minimum
- Size 2 - 12 gal. per hr. soft serve and 40 gal. per hr. shake minimum

Style A - Counter model (Type I, Size 1 or Type II, Size 1 only)
Style B - Floor model
Style C - Floor model, caster mounted

AMSC N/A FSC 4110

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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2. Salient characteristics.

2.1 Design and construction. The soft serve ice cream makers, shake makers, and combination machines shall be designed to produce and dispense soft serve ice cream and shakes at the capacities stated above in ambient temperatures of 75°F to 100°F. For this ambient condition the produced product temperature shall be 20° to 22°F for the ice cream and 26° to 28°F for shakes when using a 40°F to 45°F temperature mix. The machines shall consist essentially of a structured frame unit enclosing an air cooled condensing unit, freezer head assembly, mix tank, and all controls necessary for automatic operation. The front plate and fastening screws shall be constructed of stainless steel or high strength plastic. The exterior panels of the machines shall be fabricated of stainless steel. On all the machines, overrun and blending shall be accomplished by means of a gear or belt drive beater/dasher utilizing a gravity flow or pump system. The freezer head assembly and mix tank shall be insulated with industrial type fluorocarbon expanded polyurethane preformed to shape or foamed-in-place Foot pedal operation shall be provided in addition to the manually operated dispensing mechanism to control the product flow for each dispensing head on the floor model and caster mounted model machines, when specified (see 5.2). The floor models machines shall be equipped with dual indicator lights: One warning light is activated when the mix has reached an unsafe low level; the second indicator light is activated when the mix is exhausted and automatically shuts down the freezer. Each machine shall be provided with a compartmented tray with configuration of the component parts disassembled from the machine during the cleaning and sanitizing process for ease in identification.

2.2 Dimensions. Each machine shall not exceed the maximum dimensions specified in table I.

Table I: Maximum dimensions and models (in inches)

Type	Model	Height	Width	Depth
I	Counter	36	24	32
II	Counter	36	24	32
III	Caster	60	32	35

2.3 Freezer head section. The freezer head section for single head machines shall contain the freezing cylinder, the beater/dasher, mix tank including lid, mix feed valve, rear head, front head, and removable head door. Twin head machines shall contain two of each of the aforementioned components. The removable freezer head door shall be provided with a leakproof sealing arrangement when in a closed position. The beater/dasher shall whip air into the mix, scrape freezing mix from the surface of the cylinder, and move the product forward in the freezing cylinder to assist ejection for continuous product drawoff.

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2.4 Wash kit. The wash kits shall be furnished for type III machines and shall be ready to attach to a water supply and drain. The wash kit shall consist of, but not be limited to, a chrome plated hot and cold faucet with swivel mounting and an antisplash aerator, hot and cold copper lines, a drain hose and a stainless steel or a high strength plastic connecting device for attaching the drain hose to the serving spout or a permanently connected drain hose. The size of the water inlet supply and drain shall be the manufacturer's standard commercial model. The wash kit and drain assembly shall be designed to preclude a cross connection to the portable water supply.

2.5 Refrigeration system. The refrigeration system shall be self-contained within the frame work. The refrigeration system shall consist of an electric motor driven refrigeration compressor, air cooled condenser, controls, and other accessories necessary to operate as a completely automatic refrigeration system.

2.6 Electrical requirements. The ice cream and shake maker or combination there of, shall be 115 volt, 208/230 volt, 60 hertz, single phase or 208/230 volt, 60 hertz, three phase A.C. Controls shall be designed for operation on either 115 volt or 208/230 volt power source. Transformers, when furnished shall only be used for control system operation and shall be furnished as an integral part of the control system. When 50 hertz is specified, (see 5.2), and the end item is to be operated on that frequency, oil systems should be rated at that frequency. Motors shall conform to NEMA MG1. Cords and outlets shall conform to UL471.

2.7 Standard compliance. The soft serve ice cream maker; shake makers, and the combination machines shall comply with Underwriters Laboratories, Inc., Standard No. 621 and National Foundation Standard No. 6.

2.8 Data name plate (Government unique requirement (see 5.4)). The ice cream maker shall be furnished with a data name plate with the following information:

National Stock Number
Procurement Instrument Identification Number
Specification Data
Manufacturer's Name, Address and Telephone Number
Supplier's Name, Address and Telephone Number (List only if different from Manufacturer)
Manufacturer's Model Number
Government Approved Manual Number (See 5.2)

Each plate shall be placed so that it is readily visible to the operator during normal operating use and so as to not adversely affect the life and utility of the ice cream maker. In addition to the identification plate, a permanent type bar label shall be affixed to the machine which will be suitable for scanning by a portable bar code scanner.

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3. Certification. The contractor shall certify that the product offered meets the salient characteristics of the description and except for those government unique requirements specified, conforms to the producer's own drawings, specifications, standards and quality assurance practices, compliance with the Industry standards specified in 2.8, and is the same as the product offered for sale in the commercial marketplace. The contractor shall certify the product meets the "Government unique" requirements except when specific quality assurance provisions are specified for the Government unique requirements. In this case, the contractor shall provide data resulting from inspection conducted in accordance with the specified quality assurance. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4. PACKAGING.

4.1 Packaging and packing. Packaging and packing shall be commercial or export as specified (see 5.2).

4.2 Commercial packaging and packing. Each machine shall be cleaned, preserved, packaged, packed, and unitized in accordance with ASTM D 3951.

4.3 Export.

4.3.1 Export packaging. Each machine shall be disassembled to the extent determined by the supplier to protect parts susceptible to damage during shipment and storage and for reduction of cubage. Parts removed shall be packaged in accordance with commercial practice and be secured to the machine with pressure-sensitive reinforced tape or be placed within the machine. Parts not removed shall be secured in placed with pressure-sensitive reinforced tape. Detached legs or casters, when applicable, shall be packaged in a domestic fiberboard box constructed and closed in accordance with PPP-B-636. The machine and parts shall be cleaned, preserved, cushioned, and immobilized to afford protection against corrosion, deterioration, and physical damage during shipment and storage.

4.3.2 Export packing. Each machine shall be packed in a snug-fitting wood-cleated plywood box conforming to domestic type, style A or J, grade B, type 2 load of PPP-B-601 or nailed wood box conforming to class 1, style 2 or 4, grade B, type 2 load of PPP-B-621. The contents of each box shall be blocked and braced to prevent movement while the box is in transit. The machine shall be blocked in a manner to provide approximately 1 inch clearance between the legs or casters, when applicable, and the bottom of the box. Each shipping container shall be closed and strapped in accordance with the appendix of PPP-B-601 or PPP-B-621, as applicable.

4.4 Marking (commercial and export). Marking of unit pack, shipping container and unit load shall be in accordance with MIL-STD-129.

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5. NOTES

5.1 Intended use. The soft serve ice cream makers, shake makers, and combination ice cream and shake makers are intended for use in military dining facilities.

5.2 Ordering data.

a. Title, number, and date of this document.
b. Type required (see 1).
c. Government approved manual number to be included on data name plate (see 4.1).
d. Selection of applicable packaging and packing required (see 4.1).
e. Foot pedal operation, when specified (see 2.1).
f. When electrical characteristics other than specified are required, the contracting officer shall provide voltage, phase, cycle, and other necessary requirements, to facilitate operation and installation.

5.3 Sources of documents.

5.3.1 Sources of nongovernment association documents.

Underwriters Laboratories, Inc. (UL)

Standard No. 621 - Ice Cream Makers

(Application for copies should be addressed to Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook, IL 60062.)

National Sanitation Foundation (NSF)

Standard No. 6 - Dispensing Freezers

(Application for copies should be addressed to National Sanitation Foundation, 3475 Plymouth Road, P.O. Box 1468 Ann Arbor, MI 48106.)

National Electrical Manufacturers Association (NEMA)

(Application for copies should be addressed to National Electrical Manufacturers Association (NEMA), 2101 L Street, N.W., Washington, DC 20037.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Material, 1916 Race Street, Philadelphia, PA 19103.)

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5.3.2 Sources of government documents.

Military documents are available from:

Naval Publications and Forms Center
5801 Tabor Avenue
Philadelphia, PA 19120

5.4 Government unique requirements. Whenever "(Government unique requirements)" is included in the title of a paragraph under "salient characteristics" it is meant that the requirement is something that is not normally offered to the commercial marketplace by the manufacturer.

Custodians:

Army - GL
Navy - YD
Air Force - 99

Preparing activity:

Army - GL

Project No. 4110-0406

Review activities:

Army - MD, TS
Navy - MC, MS, SA
Air Force - 50, 82
DLA - GS

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